DATA ANALYSIS QUESTIONS

Graduation Project - DEPI Data Analysis Track

MANUFACTURING DOWNTIME ANALYSIS

1. Overview

These are the 40 business questions the Power BI dashboard must answer, in five scopes. Next to each question you see the KPI card, the chart and the fields it uses. All fields exist in the two cleaned tables, *Line productivity* (one row per batch) and *Line downtime* (one row per batch and cause), linked on Batch.

Scope	Questions	Focus
Scope 1: Production Performance Analysis	8	Production performance: batches, durations and daily pattern.
Scope 2: Downtime Analysis	8	How much time is lost, by batch, cause and day.
Scope 3: Root Cause Investigation	8	Which causes matter, operator error, and where each cause lands.
Scope 4: Production Efficiency Analysis	8	Efficiency and downtime by product and operator, outliers and correlation.
Scope 5: Predictive Analysis and Production Planning	8	Next-day downtime and the number of batches (Python forecast shown in Power BI).

- **Removed:** flavor and size questions (the data has only a product code), and duplicates (average downtime per product and average duration per operator were each asked twice).
- **Added:** operator and product by day, causes by operator, what-if efficiency, outlier batches, correlation, back-test of the forecast, and a fuller forecast scope (2 questions became 8).
- **Balanced:** 8 questions per scope instead of 8, 10, 10, 10 and 2.

2. Questions, KPI Cards and Charts

Scope 1: Production Performance Analysis (Q1 to Q8)

#	Business question	KPI card	Chart	Fields
Q1	How many batches were produced, over how many production days, and how many passed the Check?	Total Batches, Production Days	Cards	Batch, Date, Check
Q2	What is the average batch time compared with the minimum, and how much of it is run time?	Avg Duration, Avg Min batch time, Avg Run Time	Three cards	Duration, Min batch time, Run Time
Q3	Which 10 batches took the longest?	Duration (Top N)	Table with data bars	Batch, Product, Start Time, End Time, Start Hour, Duration

#	Business question	KPI card	Chart	Fields
Q4	How many batches took longer than their minimum time, and by how many minutes?	Batches over Minimum, Excess Minutes	Card and histogram	Duration, Min batch time
Q5	How did batches and production minutes vary across the four days?	Total Batches, Production Minutes by day	Column chart	Date, Batch, Duration
Q6	Which product has the longest average batch time?	Avg Duration by product	Bar chart sorted	Product, Duration
Q7	How many batches did each product and each operator run on each day?	Total Batches by day	Two stacked columns by day	Date, Product, Operator, Batch
Q8	What did each operator produce, by product (production mix)?	Total Batches	Matrix	Operator, Product, Batch

Scope 2: Downtime Analysis (Q9 to Q16)

#	Business question	KPI card	Chart	Fields
Q9	What is the total downtime and what share of production time does it take?	Total Downtime Min, Downtime %	Card and gauge	Total Downtime, Duration
Q10	How much downtime does an average batch carry, and how many batches had none?	Avg Downtime per Batch, Zero Downtime Batches	Two cards	Total Downtime, Batch
Q11	Which 10 batches lost the most time?	Total Downtime (Top N)	Table with data bars	Batch, Date, Product, Operator, Total Downtime
Q12	Which causes cost the most minutes?	Cause Minutes	Horizontal bar, sorted	Description, Minutes
Q13	What share of total downtime does each cause hold?	Cause Share %	Treemap	Description, Minutes
Q14	How long is an average stoppage for each cause?	Avg Stoppage Min	Bar chart	Description, Minutes
Q15	Which causes affected the most batches?	Batches Affected	Horizontal bar, sorted	Factor, Description, Batch
Q16	How did downtime and downtime % change from day to day?	Total Downtime, Downtime % by day	Line and column chart	Date, Total Downtime, Duration

Scope 3: Root Cause Investigation (Q17 to Q24)

#	Business question	KPI card	Chart	Fields
Q17	Which few causes together make up about 80% of the downtime?	Cumulative %	Pareto (bars and line)	Description, Minutes
Q18	How many minutes and what share of downtime is operator error?	Operator Error Min, Operator Error Share	Card and donut	Operator Error, Minutes
Q19	Which operator-error causes cost the most minutes?	Cause Minutes	Table with data bars	Description, Operator Error, Minutes
Q20	Which causes hit each product hardest?	Cause Minutes	Matrix with heat colours	Product, Description, Minutes

#	Business question	KPI card	Chart	Fields
Q21	Are some causes linked to particular operators?	Cause Minutes	Matrix with heat colours	Operator, Description, Minutes
Q22	Do batches hit by a given cause run longer than batches without it?	Avg Duration with and without the cause	Clustered bar	Description, Duration, Batch
Q23	How does each operator's downtime per batch change from day to day?	Avg Downtime per Batch by day	Line chart, one line per operator	Date, Operator, Total Downtime
Q24	What efficiency would the line reach without the top cause, and without operator error?	Efficiency without Top Cause, without Operator Error	Cards beside Overall Efficiency	Min batch time, Duration, Operator Error Minutes, Minutes

Scope 4: Production Efficiency Analysis (Q25 to Q32)

#	Business question	KPI card	Chart	Fields
Q25	What is the overall efficiency, and how does it change by day?	Overall Efficiency	Gauge and line chart	Min batch time, Duration, Efficiency, Date
Q26	Which product is the most and the least efficient?	Overall Efficiency by product	Bar chart sorted	Product, Min batch time, Duration
Q27	Which operator is the most and the least efficient?	Overall Efficiency by operator	Bar chart sorted	Operator, Min batch time, Duration
Q28	Which operator has the highest downtime per batch, and what share of it is operator error?	Avg Downtime per Batch, Operator Error Share	Clustered bar	Operator, Total Downtime, Operator Error Minutes
Q29	Which product has the highest downtime per batch?	Avg Downtime per Batch by product	Bar chart sorted	Product, Total Downtime
Q30	What share of each product's batches ran longer than the minimum?	Share of Batches over Minimum	Column chart	Product, Duration, Min batch time
Q31	Which batches had unusually high downtime (above average plus one standard deviation)?	Outlier Batches	Scatter with outliers highlighted	Batch, Total Downtime
Q32	Do products with a longer minimum batch time lose more downtime?	Correlation (Min batch time, Downtime)	Scatter plot	Product, Min batch time, Total Downtime

Scope 5: Predictive Analysis and Production Planning (Q33 to Q40)

How the forecast works: calculated in Python from the cleaned data and loaded into Power BI as a small results table. Expected batch time = minimum batch time + average downtime per batch; expected batches = operating minutes / expected batch time, rounded down. Operating minutes are an input (default 965, the daily average) chosen with a slicer.

#	Business question	KPI card	Chart	Fields
Q33	What downtime per batch should we expect on the next operating day?	Expected Downtime per Batch	Card	Total Downtime
Q34	What downtime per batch should we expect for each product?	Avg Downtime per Batch	Table	Product, Total Downtime, Batch
Q35	What is the expected time of one batch for each product?	Expected Batch Time	Table	Product, Min batch time, Total Downtime
Q36	How many batches can the line complete in the operating minutes available?	Expected Batches	Card with slicer	Operating minutes (input), Min batch time, Total Downtime
Q37	How does the batch count change in the best, average and worst case?	Expected Batches (Best, Worst)	Clustered column	Date, Total Downtime, Min batch time
Q38	How many batches are lost to downtime compared with a day without any?	Batches Lost to Downtime	Card	Min batch time, Total Downtime
Q39	Which causes should be targeted to recover the most batches?	Cause Minutes	Bar chart sorted	Description, Minutes
Q40	How accurate would the method have been on the past days?	Predicted versus actual downtime per batch	Table	Date, Total Downtime

3. Data Coverage Check

Every column of the two cleaned tables is used by at least one question.

Field	Table	Used in	Questions
Date	Line productivity	9	Q1, Q5, Q7, Q11, Q16, Q23, Q25, Q37, Q40
Product	Line productivity	12	Q3, Q6, Q7, Q8, Q11, Q20, Q26, Q29, Q30 ...
Batch	Line productivity	11	Q1, Q3, Q5, Q7, Q8, Q10, Q11, Q15, Q22 ...
Operator	Line productivity	7	Q7, Q8, Q11, Q21, Q23, Q27, Q28
Start Time	Line productivity	1	Q3
End Time	Line productivity	1	Q3
Start Hour	Line productivity	1	Q3
Duration	Line productivity	13	Q2, Q3, Q4, Q5, Q6, Q9, Q16, Q22, Q24 ...
Min batch time	Line productivity	12	Q2, Q4, Q24, Q25, Q26, Q27, Q30, Q32, Q35 ...
Total Downtime	Line productivity	16	Q9, Q10, Q11, Q16, Q23, Q28, Q29, Q31, Q32 ...
Operator Error Minutes	Line productivity	2	Q24, Q28
Run Time	Line productivity	1	Q2
Efficiency	Line productivity	1	Q25
Check	Line productivity	1	Q1
Factor	Line downtime	1	Q15
Description	Line downtime	10	Q12, Q13, Q14, Q15, Q17, Q19, Q20, Q21, Q22 ...
Minutes	Line downtime	10	Q12, Q13, Q14, Q17, Q18, Q19, Q20, Q21, Q24 ...
Operator Error	Line downtime	2	Q18, Q19

4. Definitions and Data Notes

- Duration = End Time minus Start Time in minutes (corrected for batches that cross midnight); Efficiency = Min batch time / Duration; Run Time = Duration minus Total Downtime.
- The data covers 4 production days (29, 30, 31 Aug and 2 Sep 2024), 38 batches, 6 products, 4 operators and 11 of the 12 downtime causes.
- Some products have very few batches (OR-600 has 1), so product comparisons are indicative. The data has no shift column.
- Q32 compares products, not duration with downtime, because Duration already contains the downtime.